

SKILLS SUMMARY

Data & AI Analyst passionate about uncovering insights that drive smarter storytelling and user engagement. Skilled in SQL, Python, experimentation design, and causal inference. Experienced in building analytical frameworks and dashboards that translate product data into measurable business impact. Strong background in machine learning and applied statistics with a focus on media and responsible AI.

Skills: Python, SQL, C#/ .NET | Machine Learning, NLP, Unsupervised Learning | Pandas, NumPy, scikit-learn, TensorFlow, PyTorch | UMAP, HDBSCAN | Tableau, Power BI | AWS, Azure, Docker | REST APIs | Responsible AI, Investigative Research

EDUCATION

Georgia Institute of Technology | M.S in Computer Science | Specialization in Machine Learning | 2023 - December 2025

Vanderbilt University | B.S. in Computer Science, Psychology

EXPERIENCE & LEADERSHIP

A.I. Engineer & Researcher | Emory University | November 2025 - Present

- Built an end-to-end machine learning system that transforms over ~20M words, representations for 8,000+ text segments, into **interactive visual maps**, enabling scalable exploration of themes and relationships across large text collections in conjunction with the Democracy Lab @ Emory University
- Learned semantic representations for **~8,000 text segments**, projecting high-dimensional **word embeddings** into **2D visual spaces (UMAP)** to model conceptual similarity and emergent structure in complex political and civic texts.
- Identified and validated thematic structure using **density-based clustering (HDBSCAN)** with stability checks and human-in-the-loop review, ensuring outputs were consistent, interpretable, and appropriate for public-facing research.
- Designed the system as **modular, extensible components** (data ingestion, embedding, projection, clustering, visualization) and collaborated with interdisciplinary teams to integrate ML insights into a **user-friendly web application**.

Software Engineer 1 | Cox Enterprises | April 2024 - July 2025

- Built and deployed data pipelines and analytics dashboards tracking performance across automation workflows, processing **10,000+ monthly transactions with 99% accuracy**.
- Leveraged **SQL and Python** to extract, transform, and analyze operational data, identifying key efficiency drivers that led to a **35% improvement in delivery speed** and **25% increase in adoption**.
- Collaborated with **Product, Infrastructure, and Security** teams to establish **data collection standards and reporting frameworks** for continuous monitoring and optimization.
- Implemented **robustness-focused engineering practices** (automated testing, debugging protocols, adversarial input handling, continuous monitoring) and applied a **product-led mindset** through **20+ stakeholder interviews**, improving system reliability and accelerating feature delivery.

Software Engineer | Art Sphere | Dec 2023 - Feb 2024 (Contract)

- Developed a full-stack employee management system with a dynamic front-end UI to support over 100 staff members, automating time-tracking and project info access, **saving the company \$20K+ annually and increasing user adoption by 10% and engagement by 25%**.

Cyber Security Analyst | BBS | Istanbul, Turkey | July - Aug 2022

- Enhanced system resilience by executing penetration tests and DDoS simulations, leading to a **25% faster incident response** and preventing **thousands of potential malicious requests daily** through firewall optimization.

PROJECTS & RESEARCH

Who Congress Actually Listens to on Artificial Intelligence - Built an ML-driven policy analysis pipeline for 300+ Congressional and federal AI documents, combining semantic retrieval, NER, and an influence model to quantify who shapes AI policymaking. The work revealed federal agencies held ~80% of total influence, challenging dominant media narratives. Translated findings into interactive visualizations and public-facing journalistic analysis [Who Congress Actually Listens to on Artificial Intelligence](#)

Media Bias & Public Perception of Artificial Intelligence - Analyzed 10,000+ global news headlines using Python, TextBlob, and scikit-learn to model sentiment and topics in AI coverage. Identified bias patterns across outlet ideology and editorial tone, examining how media framing and machine interpretation impact public trust and misinformation risk.

Spam Detection Model - Built and optimized ML models (Naive Bayes, Random Forest, KNN) for spam detection in large-scale Twitter datasets, focusing on improving model robustness and minimizing false positives under adversarial data conditions. **Achieved F1 score of .9 / 90% Accuracy** - [Spam Tweets.ipynb](#).